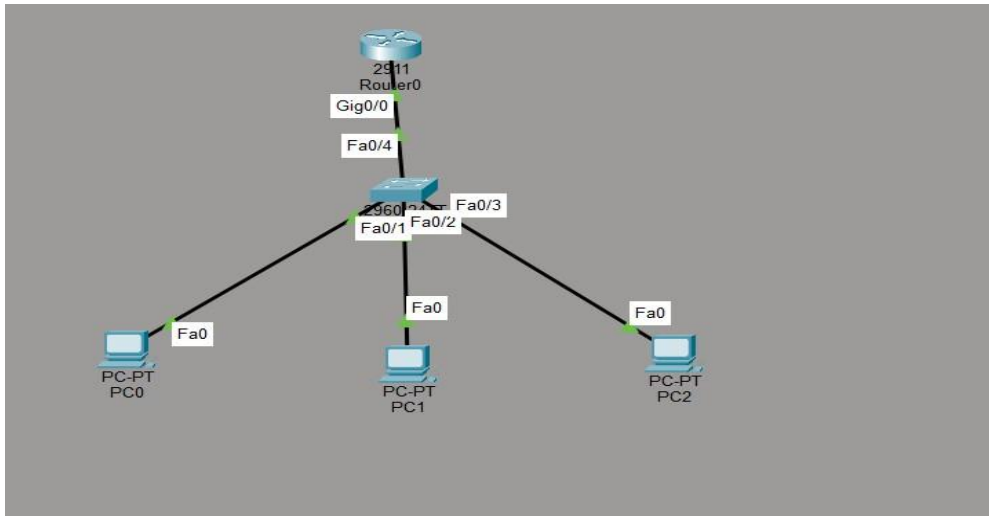


Q- Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works.

Hint: Check sub-interface IPs, trunk configuration, and default gateways on PCs.



Check in switch

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/4     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/4     10,20

Port      Vlans allowed and active in management domain
Fa0/4     10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/4     10,20

Switch#
```

Show only Vlan 10 and 20

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int f 0/4
Switch(config-if)#switchport trunk allowed vlan 10,20,30
Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
```

Add Vlan 30

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/4     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/4     10,20,30

Port      Vlans allowed and active in management domain
Fa0/4     10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/4     10,20,30

Switch#
```

Ping Test for Vlan 30 PC

```
C:\>ping 192.168.30.10
```

```
Pinging 192.168.30.10 with 32 bytes of data:
```

```
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
```

```
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 192.168.30.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>|
```